DO NOW — RECALL

Debugging and error correction

Name: _____ Class: _____ Date: _____

1. Write −1 in 8-bit two's complement.

2. What is the last index in a four-item array?

3. What precondition does binary search need?

4. What happens if code reads a[3] from a three-element C# array?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

11111111.

2. Model answer

3.

3. Model answer

Sorted input.

4. Model answer

An IndexOutOfRangeException.

Vocabulary — know these terms

off-by-one error — Loop repeats once too many or few times.

exception — Runtime failure or signal raised by an invalid operation.

trace — Record changing execution state.

precondition — Input requirement an algorithm assumes.

termination — An algorithm eventually ends.

NEW LEARNING

Read, annotate and explain

Check valid indexes

A three-element array has valid indexes 0, 1 and 2.

Follow the condition

When i is 3, i<=a.Length remains true and attempts to read a[3], causing an exception.

Fix and retest

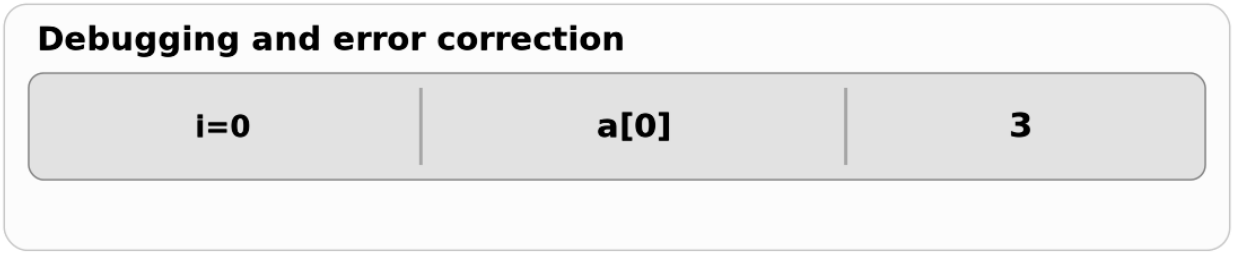
Use i<a.Length, which stops before i becomes an invalid index.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

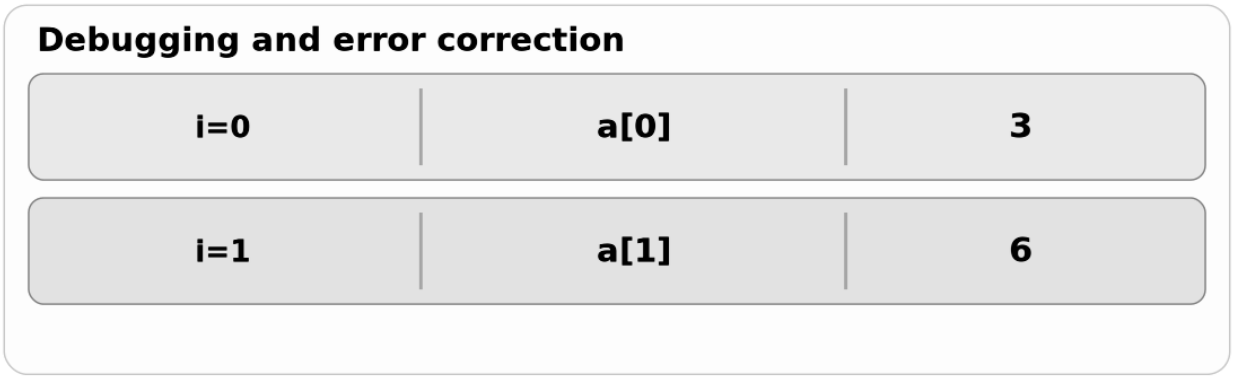
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Debugging and error correction: i=0, a[0], 3

New Learning

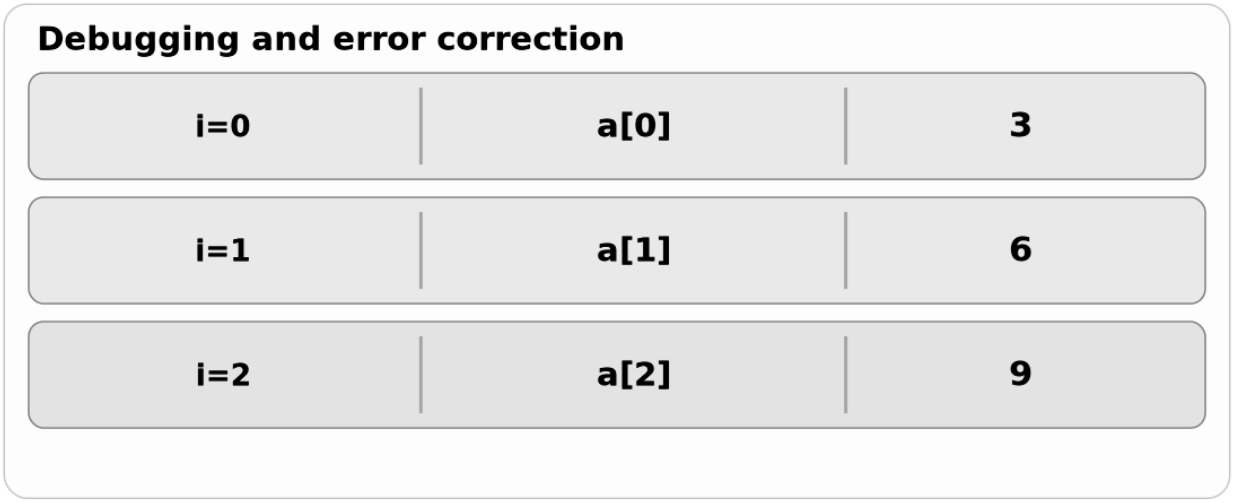


Debugging and error correction: i=0, a[0], 3; i=1, a[1], 6

New Learning · original lesson diagrams

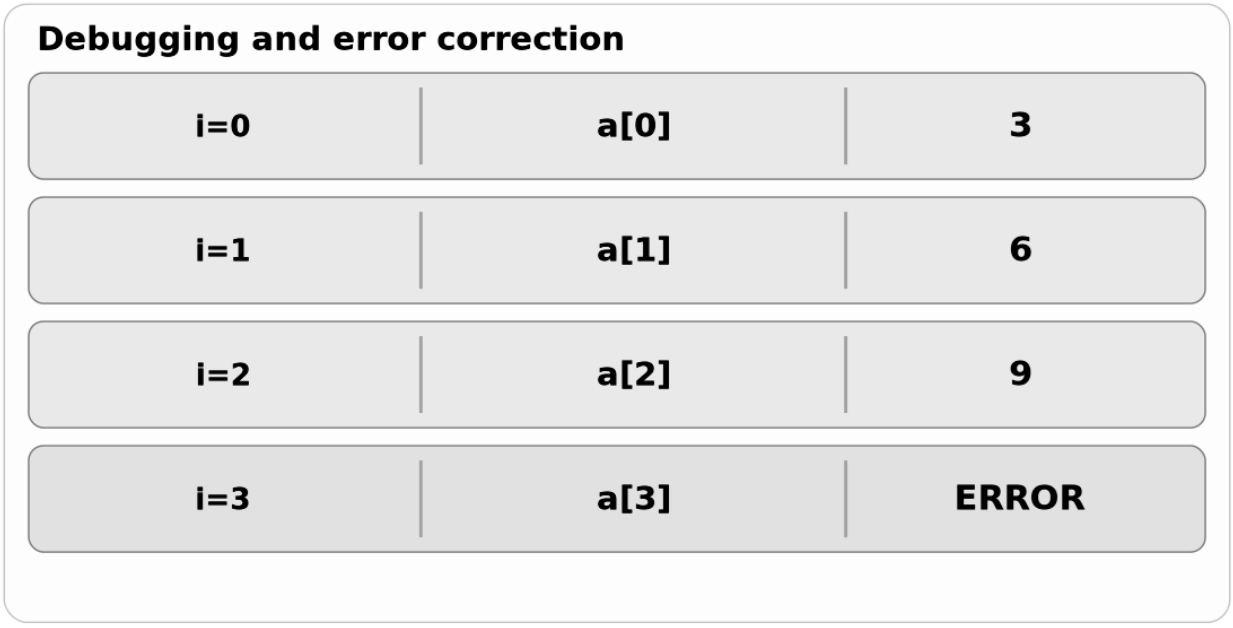
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Debugging and error correction: i=0, a[0], 3; i=1, a[1], 6; i=2, a[2], 9

New Learning



Debugging and error correction: i=0, a[0], 3; i=1, a[1], 6; i=2, a[2], 9; i=3, a[3], ERROR

I DO — WORKED EXAMPLE

Follow the worked method

Correct a loop summing [2,4] that uses <= length.

Predict

The array's valid indexes are 0 and 1, not 2.

Explain

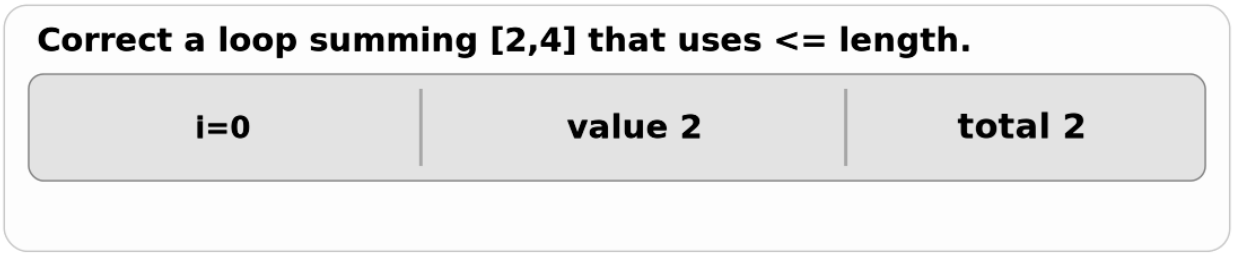
Use i < values.Length and the total is 6.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Correct a loop summing [2,4] that uses <= length.

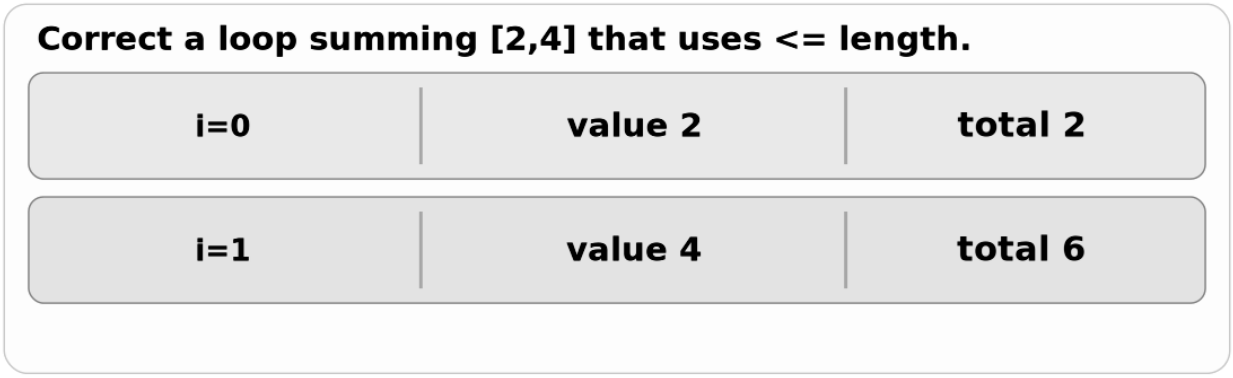
The array's valid indexes are 0 and 1, not 2.



Correct a loop summing [2,4] that uses <= length.: i=0, value 2, total 2

Correct a loop summing [2,4] that uses <= length.

The array's valid indexes are 0 and 1, not 2.



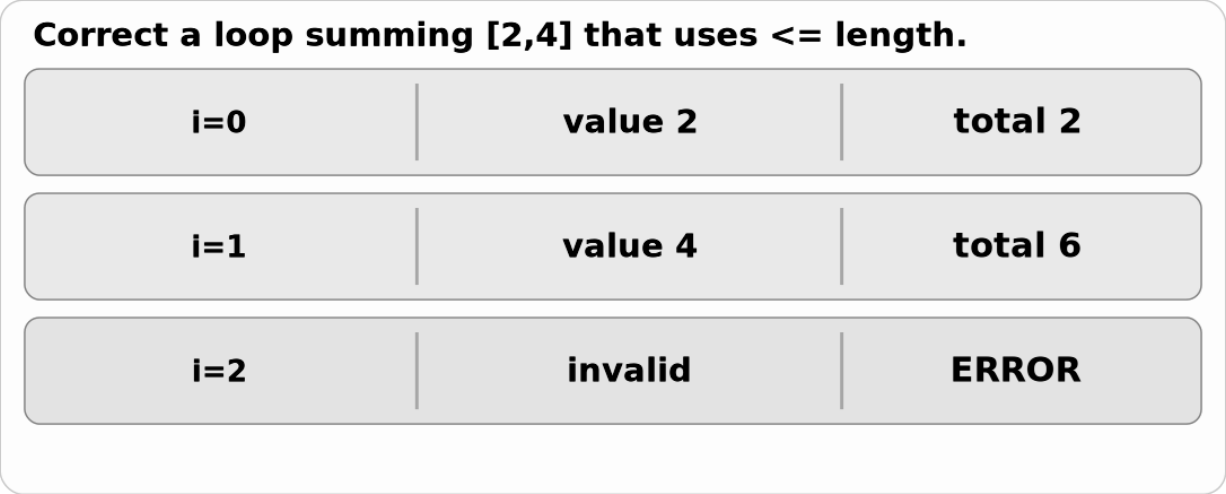
Correct a loop summing [2,4] that uses <= length.: i=0, value 2, total 2; i=1, value 4, total 6

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Correct a loop summing [2,4] that uses <= length.

The array's valid indexes are 0 and 1, not 2.



Correct a loop summing [2,4] that uses <= length.: i=0, value 2, total 2; i=1, value 4, total 6; i=2, invalid, ERROR

WE DO — GUIDED PRACTICE

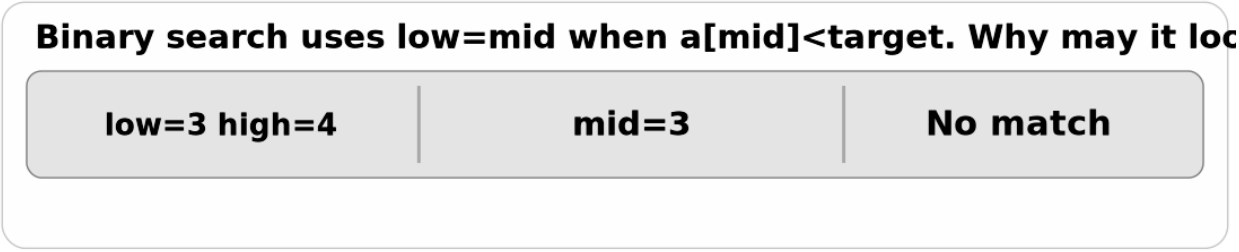
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Binary search uses low=mid when a[mid]<target. Why may it loop forever?

With low=3 and high=4, midpoint remains 3 and low=mid fails to shrink the interval.



Binary search uses low=mid when a[mid]<target. Why may it loop forever?: low=3 high=4, mid=3, No match

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict With low=3 and high=4, midpoint remains 3 and low=mid fails to shrink the interval.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Fix the loop condition i<=a.Length when visiting every array item.

Question 2

When a[mid]>target, what should binary search do to high?

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Choose normal, boundary and erroneous indexes when a method accepts only indexes 0–4.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Using low=mid+1 after a[mid]<target can prevent non-shrinking binary-search intervals.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Binary search uses low=mid when a[mid]<target. Why may it loop forever?

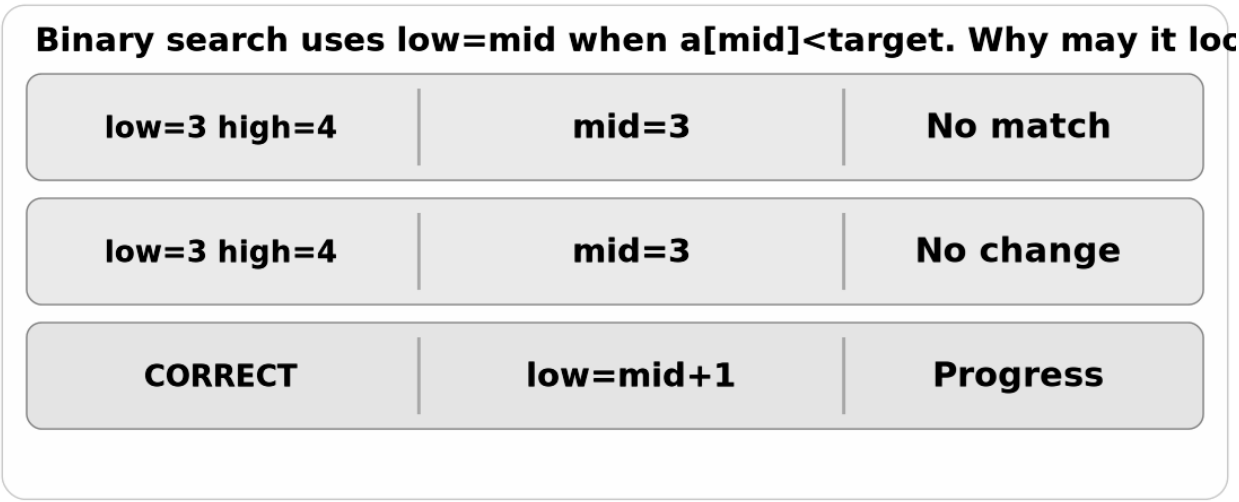
With low=3 and high=4, midpoint remains 3 and low=mid fails to shrink the interval.



Binary search uses low=mid when a[mid]<target. Why may it loop forever?: low=3 high=4, mid=3, No match; low=3 high=4, mid=3, No change

Binary search uses low=mid when a[mid]<target. Why may it loop forever?

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Binary search uses low=mid when a[mid]<target. Why may it loop forever?: low=3 high=4, mid=3, No match; low=3 high=4, mid=3, No change; CORRECT, low=mid+1, Progress

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain

Use low=mid+1 to exclude the checked midpoint and guarantee progress.

IMPROVE: Explain one correction to your own reasoning.